

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE CODE: ITA 06

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO1: Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation & Heuristic Search (BL1, BL2)

Assessment Tool Weightage: 20%

QUESTIONS: 15

Total Marks: 100

Annexure A

CONCEPT MAPPING

Code of Conduct:

I Praveen Kumar S D (192312077) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

***Signature of
the student:
Total Marks
Awarded***

<i>Criteria</i>	<i>Conceptual Understanding</i>	<i>Accuracy of Answers</i>	<i>Application of Knowledge</i>	<i>Completeness of Response</i>	<i>Clarity & Presentation</i>	<i>Total</i>
<i>Weightage</i>	30%	20%	20%	15%	15%	
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						

Annexure A

Short Answer Test

7. In the context of the **Find-S algorithm**, consider a dataset used to determine whether a loan should be approved based on attributes such as Employment Status, Salary Level, Credit Score, and Loan History. Apply the Find-S algorithm to derive the most specific hypothesis consistent with the positive examples. (5 Marks)

Dataset:

Example	Employment	Salary	Credit Score	Loan History	Loan Approved
1	Employed	High	Good	Clean	Yes
2	Employed	Medium	Good	Clean	Yes
3	Unemployed	Low	Poor	Bad	No
4	Employed	Medium	Excellent	Clean	Yes

Concept Overview

The **Find-S algorithm** finds the *maximally specific hypothesis* that fits all the **positive** examples in the training data. It ignores negative examples entirely, because Find-S assumes the target concept can be represented as a conjunction of attribute constraints, and it only needs to generalize the hypothesis just enough to keep covering the positive instances seen so far.

Each attribute in the hypothesis can take one of three forms:

- A **specific value** (e.g., "Employed") — meaning only that value is acceptable
- **"?"** — meaning any value of that attribute is acceptable
- **"∅"** — meaning no value is acceptable yet (used only in initialization)

Algorithm steps:

1. Initialize h = the most specific hypothesis: $h = \langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$
2. For each **positive** training example:
 - For each attribute in h :
 - If the attribute constraint in h is satisfied by the example, do nothing
 - Else, replace that attribute in h with the next more general constraint that is satisfied by the example (usually "?")
3. Ignore all negative examples
4. Output h as the final hypothesis

Dataset

E x	Employment	Salary	Credit Score	Loan History	Loan Approved
1	Employed	High	Good	Clean	Yes
2	Employed	Medium	Good	Clean	Yes
3	Unemployed	Low	Poor	Bad	No
4	Employed	Medium	Excellent	Clean	Yes

Step-by-Step Execution

Initialization:

$h_0 = \langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$

Example 1 (Employed, High, Good, Clean $\rightarrow$ Yes) — Positive

Since h_0 is the null hypothesis, we simply adopt this example's attribute values directly.

$h_1 = \langle \text{Employed, High, Good, Clean} \rangle$

Example 2 (Employed, Medium, Good, Clean $\rightarrow$ Yes) — Positive

Compare each attribute of the example against h_1 :

Attribute	h_1 value	Example value	Match?	Action
Employment	Employed	Employed	Yes	Keep
Salary	High	Medium	No	Generalize to "?"
Credit Score	Good	Good	Yes	Keep
Loan History	Clean	Clean	Yes	Keep

$h_2 = \langle \text{Employed, ?, Good, Clean} \rangle$

Example 3 (Unemployed, Low, Poor, Bad $\rightarrow$ No) — Negative

Find-S ignores negative examples completely. No update is made.

$h_3 = h_2 = \langle \text{Employed, ?, Good, Clean} \rangle$

Example 4 (Employed, Medium, Excellent, Clean $\rightarrow$ Yes) — Positive

Compare each attribute against h_3 :

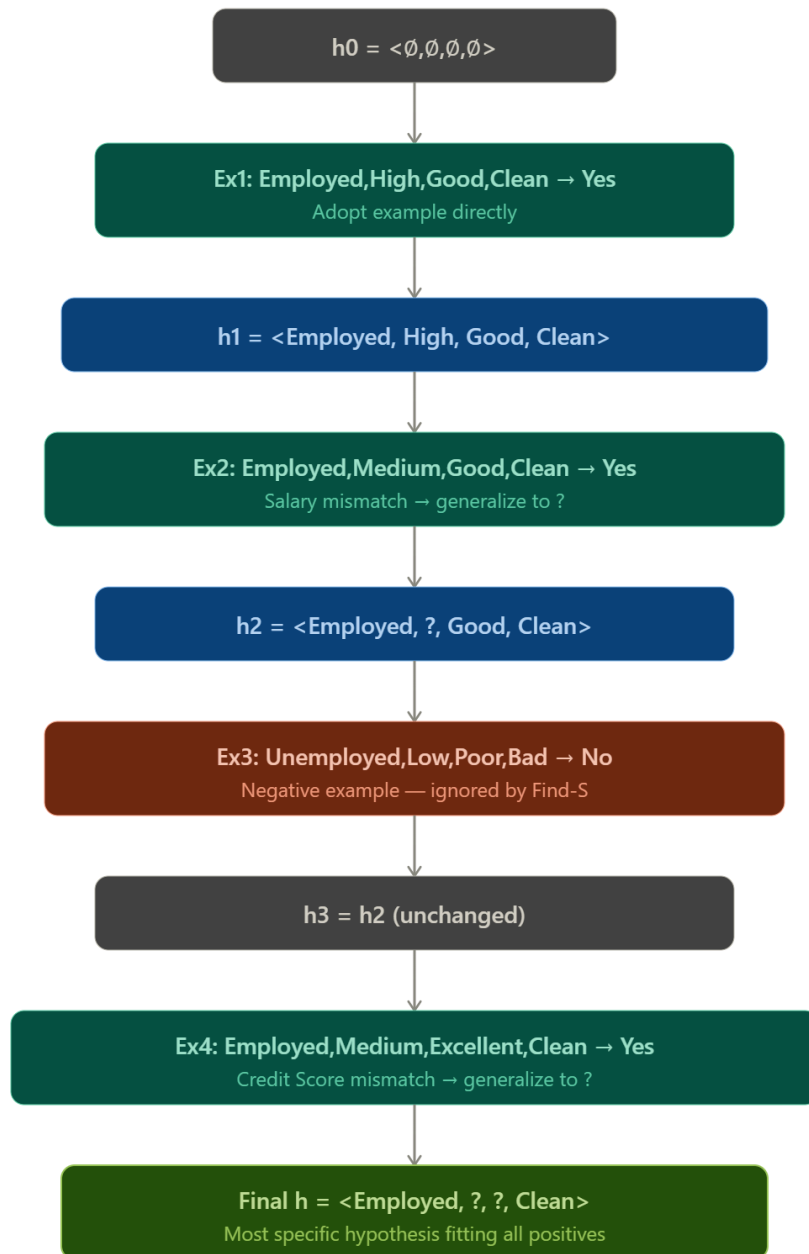
Attribute	h_3 value	Example value	Match?	Action
Employment	Employed	Employed	Yes	Keep
Salary	?	Medium	Yes (already general)	Keep
Credit Score	Good	Excellent	No	Generalize to "?"
Loan History	Clean	Clean	Yes	Keep

$h_4 = \langle \text{Employed, ?, ?, Clean} \rangle$

Final Hypothesis

$h = \langle \text{Employed, ?, ?, Clean} \rangle$

Interpretation: According to the most specific hypothesis consistent with all positive examples, a loan is approved whenever the applicant is **Employed** and has a **Clean loan history**, regardless of their Salary level or Credit Score. This makes intuitive sense from the data — every "Yes" case has Employment=Employed and Loan History=Clean, while Salary and Credit Score varied freely across the positive examples.



8. In the context of the **Candidate Elimination algorithm**, consider a dataset used to classify whether a customer will subscribe to a service based on attributes such as Age Group, Income Level, Internet Usage, and Device Type. Determine the version space by computing the **specific (S)** and **general (G)** boundaries. Show step-by-step updates after each training example. (5 Marks)

Dataset:

Example	Age Group	Income Level	Internet Usage	Device Type	Subscribe
1	Young	High	Frequent	Mobile	Yes
2	Young	High	Occasional	Mobile	Yes
3	Middle	Low	Frequent	Desktop	No
4	Young	Medium	Frequent	Mobile	Yes
5	Senior	High	Occasional	Desktop	No

Concept Overview

The **Candidate Elimination (CE) algorithm** improves on Find-S by using **both positive and negative** examples to compute the entire **version space** — the set of all hypotheses consistent with the training data — represented compactly by two boundary sets:

- **S (Specific boundary):** the set of most specific hypotheses consistent with the data
- **G (General boundary):** the set of most general hypotheses consistent with the data

Update rules:

For a **positive** example:

- Remove from G any hypothesis inconsistent with (i.e., not covering) the example
- Generalize each hypothesis in S that doesn't cover the example, minimally, so it does — but only if the generalized hypothesis is still more specific than some hypothesis in G

For a **negative** example:

- Remove from S any hypothesis that covers (incorrectly matches) the example
- Specialize each hypothesis in G that covers the example, minimally, so it doesn't — but only using attribute values actually present in S

Dataset

E x	Age Group	Income Level	Internet Usage	Device Type	Subscrib e
1	Young	High	Frequent	Mobile	Yes
2	Young	High	Occasional	Mobile	Yes
3	Middle	Low	Frequent	Desktop	No
4	Young	Medium	Frequent	Mobile	Yes
5	Senior	High	Occasional	Desktop	No

Step-by-Step Execution

Initialization:

$S_0 = \langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$

$G_0 = \langle ?, ?, ?, ? \rangle$

Example 1: Young, High, Frequent, Mobile → Yes (positive)

S is null, so it's replaced by the example directly.

- $S_1 = \langle \text{Young, High, Frequent, Mobile} \rangle$
- $G_1 = \langle ?, ?, ?, ? \rangle$ (unchanged — still consistent with the positive example)

Example 2: Young, High, Occasional, Mobile → Yes (positive)

Compare with S_1 :

Attribute	S_1	Example	Match ?	Action
Age Group	Young	Young	Yes	Keep
Income	High	High	Yes	Keep
Internet Usage	Frequent	Occasional	No	Generalize to "?"
Device	Mobile	Mobile	Yes	Keep

- $S_2 = \langle \text{Young, High, ?, Mobile} \rangle$
- $G_2 = \langle ?, ?, ?, ? \rangle$ (unchanged)

Example 3: Middle, Low, Frequent, Desktop → No (negative)

$G_2 = \langle ?, ?, ?, ? \rangle$ currently *covers* this negative example, so it must be specialized. We specialize using values from $S_2 = \langle \text{Young, High, ?, Mobile} \rangle$, attribute by attribute:

Attribute	S_2 value	Negative value	$S \neq \text{Negative?}$	New candidate G
Age Group	Young	Middle	Yes	$\langle \text{Young, ?, ?, ?} \rangle$
Income	High	Low	Yes	$\langle ?, \text{High}, ?, ? \rangle$
Internet Usage	?	Frequent	S is "?" → skip	—
Device	Mobile	Desktop	Yes	$\langle ?, ?, ?, \text{Mobile} \rangle$

- $G_3 = \{ \langle \text{Young, ?, ?, ?} \rangle, \langle ?, \text{High}, ?, ? \rangle, \langle ?, ?, ?, \text{Mobile} \rangle \}$
- $S_3 = S_2 = \langle \text{Young, High, ?, Mobile} \rangle$ (negative examples never change S)

Example 4: Young, Medium, Frequent, Mobile → Yes (positive)

Update S: Compare with $S_3 = \langle \text{Young, High, ?, Mobile} \rangle$

Attribute	S_3	Example	Match ?	Action
Age Group	Young	Young	Yes	Keep
Income	High	Medium	No	Generalize to "?"
Internet Usage	?	Frequent	Yes	Keep
Device	Mobile	Mobile	Yes	Keep

$S_4 = \langle \text{Young, ?, ?, Mobile} \rangle$

Update G: Remove any hypothesis in G_3 that does **not** cover this positive example:

- $\langle \text{Young, ?, ?, ?} \rangle \rightarrow$ covers it (Age=Young ✓) → **keep**
- $\langle ?, \text{High}, ?, ? \rangle \rightarrow$ Income=Medium $\neq$ High → does **not** cover → **remove**

- $\langle ?, ?, ?, \text{Mobile} \rangle \rightarrow \text{covers it (Device=Mobile } \checkmark) \rightarrow \text{keep}$
- $G_4 = \{ \langle \text{Young}, ?, ?, ? \rangle, \langle ?, ?, ?, \text{Mobile} \rangle \}$

Example 5: Senior, High, Occasional, Desktop $\rightarrow$ No (negative)

Check whether any hypothesis in G_4 covers this negative example:

- $\langle \text{Young}, ?, ?, ? \rangle \rightarrow \text{Age=Senior} \neq \text{Young} \rightarrow \text{does not cover negative} \rightarrow \text{already safe, no change}$
- $\langle ?, ?, ?, \text{Mobile} \rangle \rightarrow \text{Device=Desktop} \neq \text{Mobile} \rightarrow \text{does not cover negative} \rightarrow \text{already safe, no change}$

$$G_5 = G_4 = \{ \langle \text{Young}, ?, ?, ? \rangle, \langle ?, ?, ?, \text{Mobile} \rangle \}$$

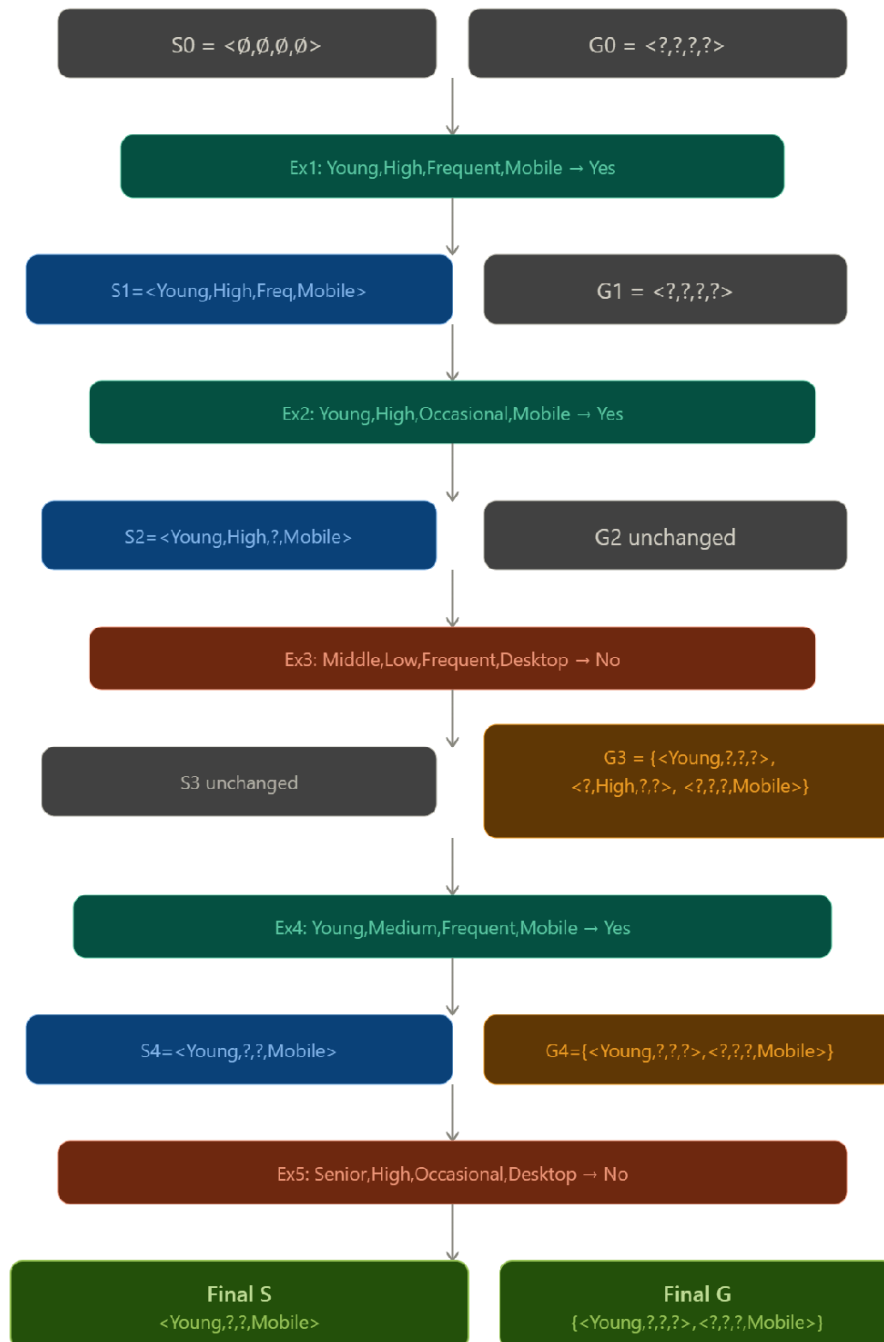
$$S_5 = S_4 = \langle \text{Young}, ?, ?, \text{Mobile} \rangle$$

Final Version Space

$$S = \{ \langle \text{Young}, ?, ?, \text{Mobile} \rangle \}$$

$$G = \{ \langle \text{Young}, ?, ?, ? \rangle, \langle ?, ?, ?, \text{Mobile} \rangle \}$$

Interpretation: Every hypothesis between S and G in the partial-order lattice is consistent with the data. The most specific rule requires *both* Age=Young *and* Device=Mobile; the most general boundary shows that *either* condition alone (Young age, or Mobile device) is sufficient to explain the observed positive/negative split.



9. In the context of the **Candidate Elimination algorithm**, consider a dataset used to classify whether a patient follows a healthy lifestyle based on attributes such as Diet, Exercise, Sleep, and Stress Level. Apply the algorithm to compute the **version space (S and G boundaries)** and show the updates after each example. (5 Marks)

Dataset:

Example	Diet	Exercise	Sleep	Stress Level	Healthy Lifestyle
1	Balanced	Regular	Adequate	Low	Yes
2	Balanced	Regular	Poor	Medium	Yes
3	Junk	None	Poor	High	No
4	Balanced	Moderate	Adequate	Low	Yes
5	Junk	Regular	Adequate	High	No

Step-by-Step Execution

Initialization:

$S_0 = \langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$

$G_0 = \langle ?, ?, ?, ? \rangle$

Example 1: Balanced, Regular, Adequate, Low → Yes (positive)

S is null, adopt the example directly.

- $S_1 = \langle \text{Balanced, Regular, Adequate, Low} \rangle$
- $G_1 = \langle ?, ?, ?, ? \rangle$

Example 2: Balanced, Regular, Poor, Medium → Yes (positive)

Attribute	S_1	Example	Match?	Action
Diet	Balanced	Balanced	Yes	Keep
Exercise	Regular	Regular	Yes	Keep
Sleep	Adequate	Poor	No	Generalize to "?"
Stress	Low	Medium	No	Generalize to "?"

- $S_2 = \langle \text{Balanced, Regular, ?, ?} \rangle$
- $G_2 = \langle ?, ?, ?, ? \rangle$ (unchanged)

Example 3: Junk, None, Poor, High → No (negative)

$G_2 = \langle ?, ?, ?, ? \rangle$ covers this negative example, so it must be specialized using values from $S_2 = \langle \text{Balanced, Regular, ?, ?} \rangle$:

Attribute	S_2 value	Negative value	$S \neq \text{Negative?}$	New candidate G
Diet	Balanced	Junk	Yes	$\langle \text{Balanced, ?, ?, ?} \rangle$
Exercise	Regular	None	Yes	$\langle ?, \text{Regular, ?, ?} \rangle$
Sleep	?	Poor	S is "?" → skip	—

Attribute	S ₂ value	Negative value	S ≠ Negative?	New candidate G
Stress	?	High	S is "?" → skip	—

- $G_3 = \{ \langle \text{Balanced}, ?, ?, ? \rangle, \langle ?, \text{Regular}, ?, ? \rangle \}$
- $S_3 = S_2 = \langle \text{Balanced}, \text{Regular}, ?, ? \rangle$

Example 4: Balanced, Moderate, Adequate, Low → Yes (positive)

Update S: Compare with $S_3 = \langle \text{Balanced}, \text{Regular}, ?, ? \rangle$

Attribute	S ₃	Example	Match?	Action
Diet	Balanced	Balanced	Yes	Keep
Exercise	Regular	Moderate	No	Generalize to "?"
Sleep	?	Adequate	Yes	Keep
Stress	?	Low	Yes	Keep

$S_4 = \langle \text{Balanced}, ?, ?, ? \rangle$

Update G: Remove any hypothesis in G_3 that does **not** cover this positive example:

- $\langle \text{Balanced}, ?, ?, ? \rangle \rightarrow$ covers it (Diet=Balanced ✓) → **keep**
- $\langle ?, \text{Regular}, ?, ? \rangle \rightarrow$ Exercise=Moderate $\neq$ Regular → does not cover → **remove**

$G_4 = \{ \langle \text{Balanced}, ?, ?, ? \rangle \}$

Example 5: Junk, Regular, Adequate, High → No (negative)

Check whether G_4 's hypothesis covers this negative example:

- $\langle \text{Balanced}, ?, ?, ? \rangle \rightarrow$ Diet=Junk $\neq$ Balanced → already does not cover the negative → **no change needed**

Also check S_4 against the negative example (it must *not* cover it, as a consistency check):

- $S_4 = \langle \text{Balanced}, ?, ?, ? \rangle \rightarrow$ Diet=Junk $\neq$ Balanced → does not cover it either → consistent, no removal needed.

$G_5 = G_4 = \{ \langle \text{Balanced}, ?, ?, ? \rangle \}$

$S_5 = S_4 = \langle \text{Balanced}, ?, ?, ? \rangle$

Final Version Space

$S = \{ \langle \text{Balanced}, ?, ?, ? \rangle \}$

$G = \{ \langle \text{Balanced}, ?, ?, ? \rangle \}$

Interpretation: Here S and G have **converged to the same single hypothesis**, meaning the version space contains exactly one hypothesis and the algorithm has found the exact target concept consistent with all 5 examples:

